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Stable tooth contacts in intercuspal occlusion makes for utilities of the jaw elevators during maximal voluntary clenching

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SUMMARY **Data are inconsistent concerning whether the level of the surface electromyographic (SEMG) activity of jaw-closing muscles increases when biting forces elevated during maximal voluntary clenching (MVC). In this study, T-Scan III system and BioEMG III system were used to record bite force, occlusal contacts and SEMG activity of the anterior temporalis (TA) and of the masseter muscles (MM) simultaneously. Recordings were obtained from 16 healthy young adult males during different conditions: (i) a fast MVC from resting position to intercuspal position (ICP); (ii) mandibular movements from ICP to protrusive or lateral edge-to-edge positions with teeth in contact with biting; (iii) a fast MVC in protrusive and lateral edge-to-edge positions. A higher level of SEMG activity was associated with a higher bite force during occluding movements ($P < 0.05$). However, during fast MVC from rest to ICP, the largest number of occlusal contacts was achieved and distributed more symmetrically, the highest level of biting force was obtained, but the SEMG activity of the jaw elevator muscles was reduced compared with its maximum level ($P < 0.05$). This phenomenon was not observed during the fast MVC in protrusive or lateral edge-to-edge positions. The present results that a lower SEMG activity was associated with the largest number of occlusal contacts and the highest level of bite force during centric MVC demonstrated a complex integration of jaw-closing muscles when a stable occlusion is present.**

KEYWORDS: **occlusal contact, bite force, jaw-closing muscle, T-Scan, electromyography**

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Introduction

Previous investigations have reported that electromyographic (EMG) activity of jaw-closing muscles during clenching is determined predictably by the level of bite force (1–3), jaw position (4), the location (5) and number of occlusal contacts (6, 7) with different responses from anterior temporalis and masseter muscles (7–11). Subjects with a larger number of occlusal contacts showed higher muscular EMG activity (4) with a linear relationship between the elevator EMG values and the bite force (12). These observations should be related to the higher number of posterior contacts, which give stable intercuspal contacts to allow elevator muscles to achieve higher activity during clenching (10, 11). The term 'stable occlusion' is defined in prosthodontics as 'the equalization of tooth contacts that prevents tooth movement after closure' (13), although in orthodontics, it includes the relapse status of a finished orthodontic occlusion, which is stable to prevent relapse and unstable in relapse cases (14–16). Baba, *et al.* (2, 4) described that with artificial premature contacts, there is a reduction in muscular activity in anterior and posterior temporal muscles on the working side and an increase on the non-working side. Regional loss of

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tooth contacts or a unilateral premature contact or interference can affect jaw movement (17, 18) with an unpredictable (complex) pattern of the jaw-closing activity (1). However, most of these previous studies were obtained by or concluded from separate recordings of the occlusal force, tooth contacts and EMG activity of jaw-closing muscles.

T-Scan system was designed to record the distribution of tooth contacts and biting force with time. It can be recorded simultaneously with EMG activity. With this combined system, the separate diagnostic data of occlusal force, tooth contacts and EMG can be recorded and played back simultaneously. The resultant playback of the recorded data illustrates the incremental analysis of the occlusal impact on muscle function. Kerstein, *et al.* (6, 19), based on simultaneous recordings of occlusal contact with T-Scan and surface electromyographic (SEMG) activity from bilateral anterior temporalis and masseter muscles of patients with myofacial pain, suggested disclusion time indicated by the T-Scan system as a parameter to evaluate the functional improvement. This was based on data indicating that the prolonged disclusion time was shorter and the contractile activity of the masseter and temporalis muscles was reduced following treatment.

Given the need for systematic studies that involve defined occlusal contact and jaw-closing muscle activity, the aim of this study was simultaneously recording of bite force, occlusal contacts and SEMG activity from the jaw-closing muscles. This was to investigate whether the number of occlusal contacts was linked with the highest level of bite force during centric maximal voluntary clenching (MVC) and whether the SEMG in anterior temporalis and the masseter muscles reached to the maximum levels. The hypothesis that during MVC in ICP when the bite force increased to the highest level, the number of occlusal contacts would increase bilaterally with the highest SEMG activity in jaw-closing muscles.

Materials and methods

Subjects

Sixteen healthy males (mean age 23·1 ± 1·5 years) of 45 volunteers participated this study. All participants presented with at least 28 permanent teeth, bilateral Class I molar relationship, an optimal 2- to 3-mm overbite and overjet, no signs or symptoms of temporomandibular disorders or known bruxism, no orthodontic appliances or history of orthognathic surgery, and healthy periodontal tissues (the probing depths were <3 mm and no clinical evidence of periodontal bone loss) (20). No subject had facial asymmetry. Two experienced clinicians (XR and Y) performed the examinations. Agreement of both examiners was required for subject selection. Those subjects that could not achieve agreement between the two examiners were excluded. All 16 volunteers who satisfied the including criteria signed an informed consent, and the study was approved by the local ethical committee of the Fourth Military Medical University.

Each subject was comfortably seated with back support and feet flat on the floor, with their eyes directed at a mark 2 m away at eye level, and their head positioned with the Frankfort horizontal plane parallel to the floor, and they were encouraged verbally to clench with maximal effort in the tests.

Testing tasks

Each subject performed eight tasks in a randomised order. (1) To record SEMG values during a 10-s period of MVC at ICP for calculating the normalised values of SEMG activity during other tasks. (2) to (5) MVC as fast as possible from rest to ICP and from light contact to MVC at protrusive edge-to-edge contact position (Pro) and at left or right edge-to-edge position defined as when the tips of the upper and lower buccal cusps of molars, premolars and canine are aligned (LP or RP). (6) to (8) Protrusive movement from ICP with heavy occlusal contact, stopping at anterior edge-to-edge position then MVC at Pro (ICP-Pro) and left or right lateral excursive movement from ICP with heavy occlusal contact, stopping at left or right edge-to-edge position then MVC at LP or RP (ICP-LP or ICP-RP).

The jaw posture tasks were demonstrated and explained to each subject using dental casts. Then, all individuals, while looking into the mirror, repeated the jaw tasks at least five times prior to the recordings. The initial and final occlusal positions were visually checked by the examiners during recordings. Each task performed thrice and each recording was lasted for 10 s. Two-minute intervals were allowed between each recording to avoid muscle fatigue. The

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interval between tasks was 5 min, and the total recording time was about 50 min.

Recordings of occlusal contacts, biting force and SEMG

The occlusal contacts and bite force were recorded with T-Scan III occlusal analysis system*, which consists of a sensor and a support, the handle assembly, the processing unit and software (6, 19). SEMG is recorded with a BioEMG III electromyographic recording system† and performed with surface electrodes‡, generally self-adhesive and positioned over the palpated main bulk of the contracted muscles. The electrodes are oriented parallel to the general direction of the centre of anterior fibres of temporalis (TA) and masseter muscles (MM). Sampling rate was set at 1000 samples per second, and the sampling time was 10 s. A common reference electrode was placed on the back of the neck, over the 7th cervical vertebral segment (10, 11). The SEMG activity and occlusal contacts during each task were recorded synchronously with the aid of T-Scan/BioEMG linking software§. All subjects were asked to perform MVC or heavy grinding movements on the sensors of T-Scan for each task mentioned above.

Other parameters

For each recording, the following parameters are provided automatically by the software of the T-Scan III system (Tekscan, Inc.). (i) Number of the occlusal contacts on different sections of the arches. Left and right halves of the arch was identified for the lateral excursion tasks and as anterior and posterior components at the mesial margin of the canine for protrusion. (ii) Spatial distribution as a percentage of biting force in different dental sections. This parameter describes the distributing percentage of the bite force, for example, on left or right, or anterior and posterior zones. (iii) The relative force base on the value of the maximum force on the whole arch provided by the T-Scan system for each task and the relative force is the percentage value of an interesting time point to this maximum force value of the corresponding task. (iv) The centre of bite force (COF), this parameter describes the coordinate values of the centre of the resultant occlusal force on the whole arch in x- and y-axes during clenching to illustrate the balance of the force sustained by the mandibular dentition (Fig. 1).

Sampling of recordings

In the fast MVC tasks, each record was sampled for 3 s from the middle section of the whole 10-s data. The mean of each parameter during each task from each recording, including the number of occlusal contacts, the relative force, the coordinate values of COF and SEMG values, was used for statistical analysis. During fast MVC in ICP/Pro/LP/RP, the number of occlusal contacts and SEMG values were selected for statistical analysis from three time points according to the bite force level. The time points were when the relative force was 25%, 50% and 100% of the maximal force, recorded, for those during fast MVC in ICP, as 25%-ICP, 50%-ICP and ICP-MVC, respectively. In the case that no point could be located on the graphs that matched the exact values of 25% or 50% relative force, the point closest to the corresponding value was selected. Finally, the selected minimum points were as follows: 19·6% and 45·4% for 25%-ICP and 50%-ICP and maximum were 27·8% and 54·2% for 25%-ICP and 50%-ICP, respectively. In protrusive

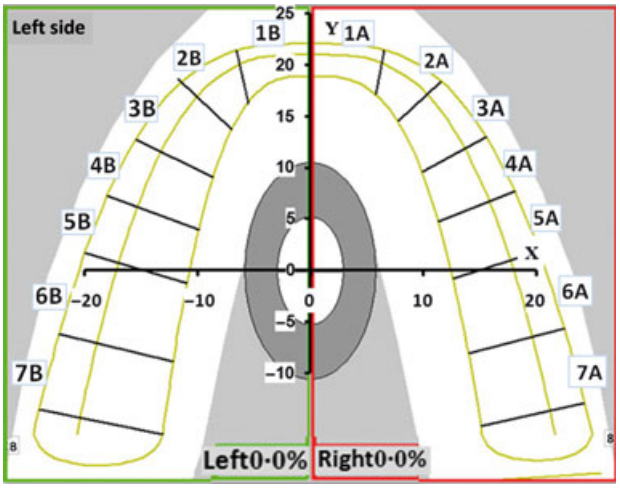

Fig. 1. An example of T-Scan system displayed on the screen. The x- and y-axes are software programmed based on the cross-hair and the centre target.

*Tekscan, Inc., Boston, MA, USA.

†Bioresearch Associates, Inc., Milwaukee, WI, USA.

‡BioFLEX: Bioresearch Associates, Inc., Milwaukee, WI, USA.

§Tekscan Inc/Bioresearch Associates technology partnership.

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and lateral excursive movements, in addition to the starting point, the ICP-MVC, and the time points of 25%-MVC and 50%-MVC, the time points of minimal and maximal force at edge-to-edge positions were also recorded, the latter two were recorded as Pro/L/R-Min and Pro/L/R-Max, respectively. Each sampled SEMG value provided by the BioEMG III system (Bioresearch Associates, Inc.) was normalised to the value of MVC in ICP for each muscle according to the following formula as previously reported (10, 11).

$$\text{Normalised SEMG value} = \frac{\textit{each sampled normalised SEMG value}}{\textit{SEMG recorded during MVC in ICP}}$$

Statistical analysis

The SPSS 13.0 package[¶] was used to describe and analyse the data. (1) Student's *t*-test was used to compare side differences. Because no significant side differences were found in these healthy volunteers, the data from both sides were pooled for further analysis. (2) Student's *t*-test was used to compare occlusal contact and force level data between anterior and posterior sections and to compare the SEMG values between working and balancing sides during static tasks. (3) One-way analysis of the variance (ANOVA) models was used to test the effects of different static tasks. Student–Newman–Keuls post-test was performed if significances between the groups were found. (4) Repetitive measurement and analysis of ANOVA were used to test the effects of time points during protrusive and lateral eccentric movements. The level of significance was set at $P < 0{\cdot}05$ for all statistical tests.

Results

Changes in bite force level, occlusal contact number and SEMG activity during centric fast MVC

The time for the total occlusal force increasing from 25%-ICP to ICP-MVC was 1·55 ± 0·84 s. The percentage of the bite force gradually balanced between left and right halves of the arches during the processing of ICP-MVC (Fig. 2a). There was a decrease in the occlusal force in the anterior section of the arch but an increase in the posterior sections of the arch ($P < 0{\cdot}05$) (Fig. 2b). The occlusal contacts were mainly distributed in posterior sections. The number of occlusal contacts gradually increased in both halves of the dental arches, especially in posterior sections as the bite force increased ($P < 0{\cdot}05$) (Fig. 2c,d). Interestingly, the SEMG activity of elevators showed no significant changes from 25%-ICP to 50%-ICP ($P > 0{\cdot}05$), but reduced from 50%-ICP to ICP-MVC ($P < 0{\cdot}05$), although the reduction in left anterior temporalis (LTA) did not reach statistical significance ($P = 0{\cdot}741$) (Fig. 2e). The SEMG values during MVC in ICP were 142·3 ± 61·1 μv and 116·4 ± 43·1 μv for LTA and right anterior temporalis (RTA), respectively, and 150·2 ± 74·5 μv and 152·2 ± 91·6 μv for left masseter muscle (LMM) and right masseter muscle (RMM), respectively, without significant side differences ($P > 0{\cdot}05$).

Alteration of force distribution during eccentric biting movement from ICP

With MVC in ICP, posterior teeth sustained the higher percentage of bite force ($P < 0{\cdot}001$) (Fig. 3a) without significant difference between left and right ($P > 0{\cdot}05$) (Fig. 3b). The COF was located near the midline at the horizontal level of the midpoint between the first molars. With MVC in Pro, LP or RP, the bite force was sustained by anterior teeth or working side teeth (Fig. 3a,b) and the COF located near the midline at the lingual side of the mesial margin of the central incisors during anterior MVC, but near the distal margin of the left or right canine during lateral MVC (Fig. 3c).

During protrusive and lateral eccentric movements, the relative force, which is defined as the percentage value of the force at the interesting time point (s) to the maximum value of the task, decreased for MVC in ICP-Pro/LP/RP to Pro/L/R-Min, but increased from Pro/L/R-Min to Pro/L/R-Max ($P < 0{\cdot}001$)(Fig. 3d). From ICP-MVC to Pro, the percentage of the bite force, in both anterior and posterior sections, did not change statistically before 50%-MVC ($P > 0{\cdot}05$) (Fig. 3d), but increased in the anterior section and decreased in the posterior section from 50%-MVC to Pro-Min ($P < 0{\cdot}05$), and was unchanged between Pro-Min and Pro-Max ($P > 0{\cdot}05$)(Fig. 3e). From ICP-MVC

¶SPSS Co., Chicago, IL, USA.

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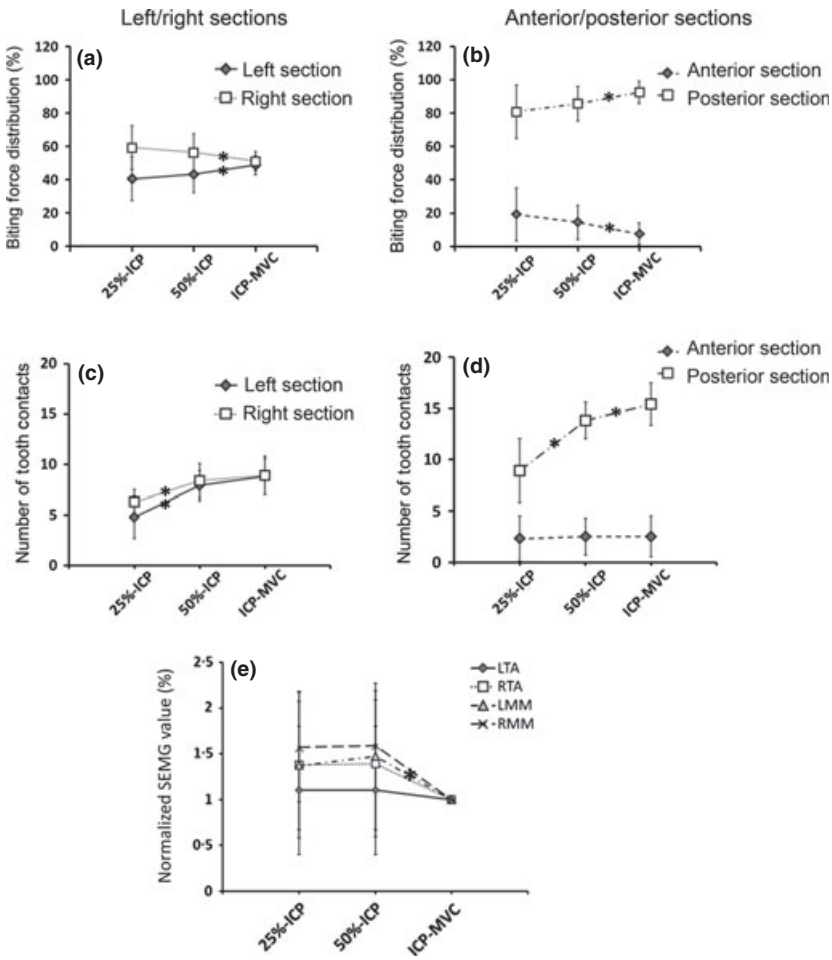

Fig. 2. Changes in the mean (± s.d.), the bite force percentage distribution (a, b) and number of occlusal contacts (c, d) in the left and right sections (a, c) and the anterior and posterior sections (b, d) of the arches and the normalised values (mean ± s.d.) of the surface electromyographic (SEMG) activity of anterior temporalis and masseter muscles (e) from light clenching to the maximum voluntary clenching (MVC) in ICP. Key: ICP-MVC = MVC in ICP, 25%-ICP = the occlusal force of around 25% of ICP-MVC, 50%-ICP = the occlusal force of around 50% of ICP-MVC, LTA = left anterior temporalis, RTA = right anterior temporalis, LMM = left masseter muscle, RMM = right masseter muscle. *$P < 0{\cdot}05$.

to the L/R-Min, there was a significant increase in the percentage of bite force on the working-side, but a decrease on the balancing-side ($P < 0{\cdot}001$), and remained unchanged between Pro-Min and Pro-Max ($P > 0{\cdot}05$) (Fig. 3f).

Alteration of occlusal contact number and distribution during eccentric biting movement from ICP

During MVC in Pro, there were no differences in occlusal contact number between left and right halves of the arch as that in ICP ($P > 0{\cdot}05$). The occlusal contacts were mainly posteriorly in ICP ($P < 0{\cdot}001$), but not posteriorly in Pro ($P < 0{\cdot}001$). During MVC in LP or RP, occlusal contacts were located only on the working side but were fewer than in ICP ($P < 0{\cdot}001$) (Fig. 4a,b).

During ICP to Pro, the number of occlusal contacts gradually reduced posteriorly ($P < 0{\cdot}001$) but increased anteriorly from 25%-MVC to Pro-Min ($P < 0{\cdot}05$) and was similar from Pro-Min to Pro-Max ($P > 0{\cdot}05$) (Fig. 4c). During ICP-LP or ICP-RP, there were no differences in the number of occlusal contacts between ICP-MVC and 50%-MVC on the working-side ($P > 0{\cdot}05$), but there was a significant decrease on the balancing-side ($P < 0{\cdot}001$). And from 50%-MVC to L/R-Min, the number of occlusal contacts gradually decreased on both working-side and balancing-sides ($P < 0{\cdot}001$) and was identical between Pro-Min and Pro-Max ($P > 0{\cdot}05$) (Fig. 4d).

Alteration of the SEMG values during eccentric biting movement from ICP

The normalised SEMG activity in Pro showed no side differences ($P > 0{\cdot}05$), while in tasks LP or RP, a higher SEMG activity was found in ipsilateral TA and contralateral MM compared with their opposite muscles (all $P < 0{\cdot}05$). No significant changes were noticed in SEMG activity for bilateral TA and MM

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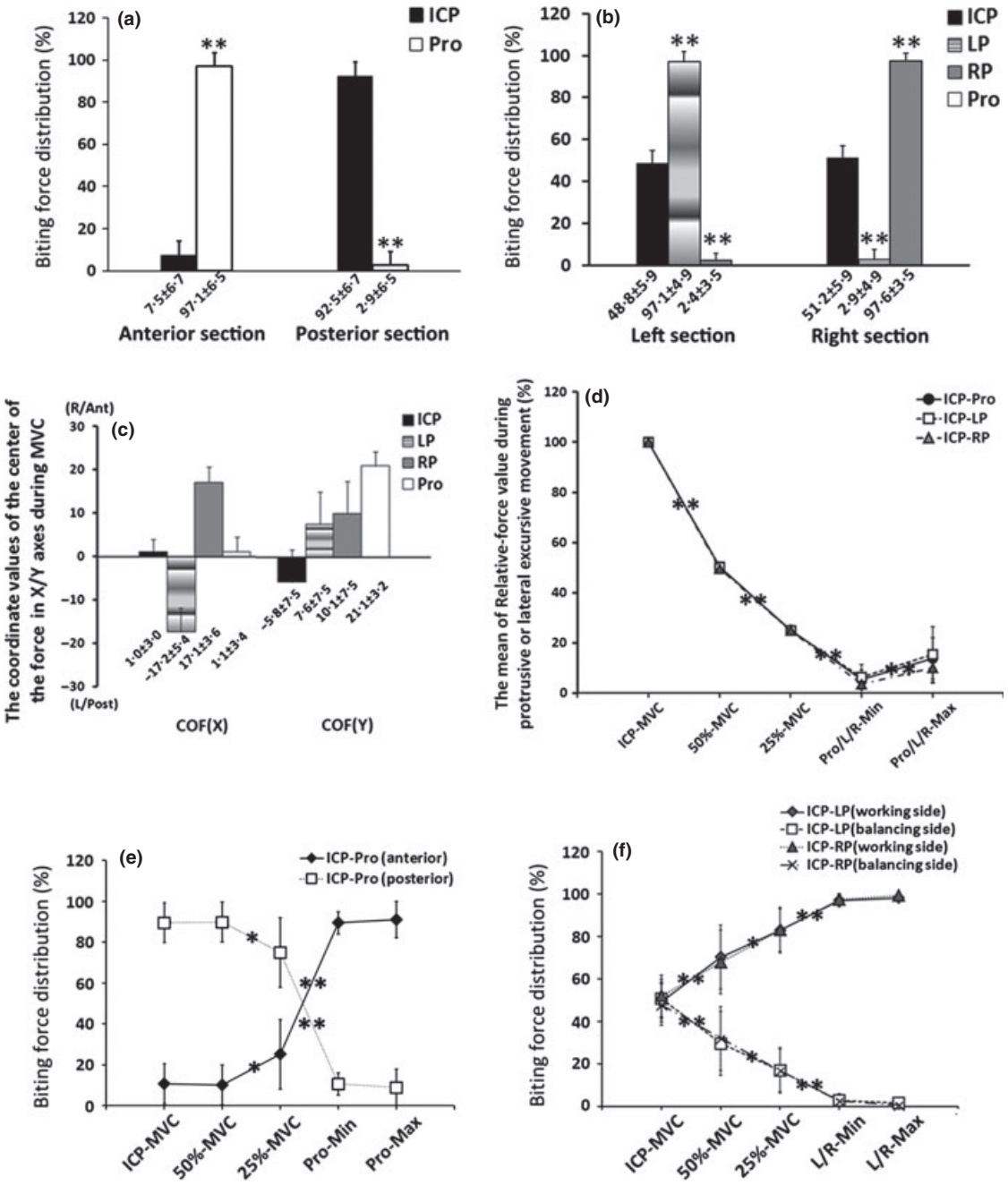

Fig. 3. The mean (± s.d.) of bite force distribution in percentage of anterior and posterior sections (a) and in left and right sections (b) of the arch and the mean (± s.d.) of coordinate values of the centre of bite force (COF) in x (left–right)- and y (anterio-posterior)-axes (c) during maximal voluntary clenching (MCV) in ICP, Pro, LP or RP. Changes in mean (± s.d.) of the relative force value of the entire arch (d) and in the biting force distribution in percentage of anterior and posterior sections (e) and in left and right sections (f) of the arch during protrusive and lateral excursive movements with tooth contact. Key: ICP = maximum intercuspal position; Pro = protrusive edge-to-edge contact position; LP/RP = left or right edge-to-edge contact position. ICP-Pro/LP/RP = movement from ICP to anterior or left/right lateral edge-to-edge positions with tooth in contacts. ICP/50%/25%-MVC = the position when the occlusal force is 100%/50%/25% of that maximal voluntary clenching (MVC) in ICP during protrusive or lateral excursive movement from ICP. Pro/L/R-Min/Max = minimal/maximal biting in protrusive or left/right lateral edge-to-edge contact position during protrusive or lateral excursive movement from ICP. The numbers at the bottom in the x-axis of 'a' and 'b' mean the values of the biting force percentage distributing in the different sections and in the *x*-axis of 'c' mean the coordinate values of the centre of force in x-/y-axes. $*P < 0{\cdot}05$, $**P < 0{\cdot}01$.

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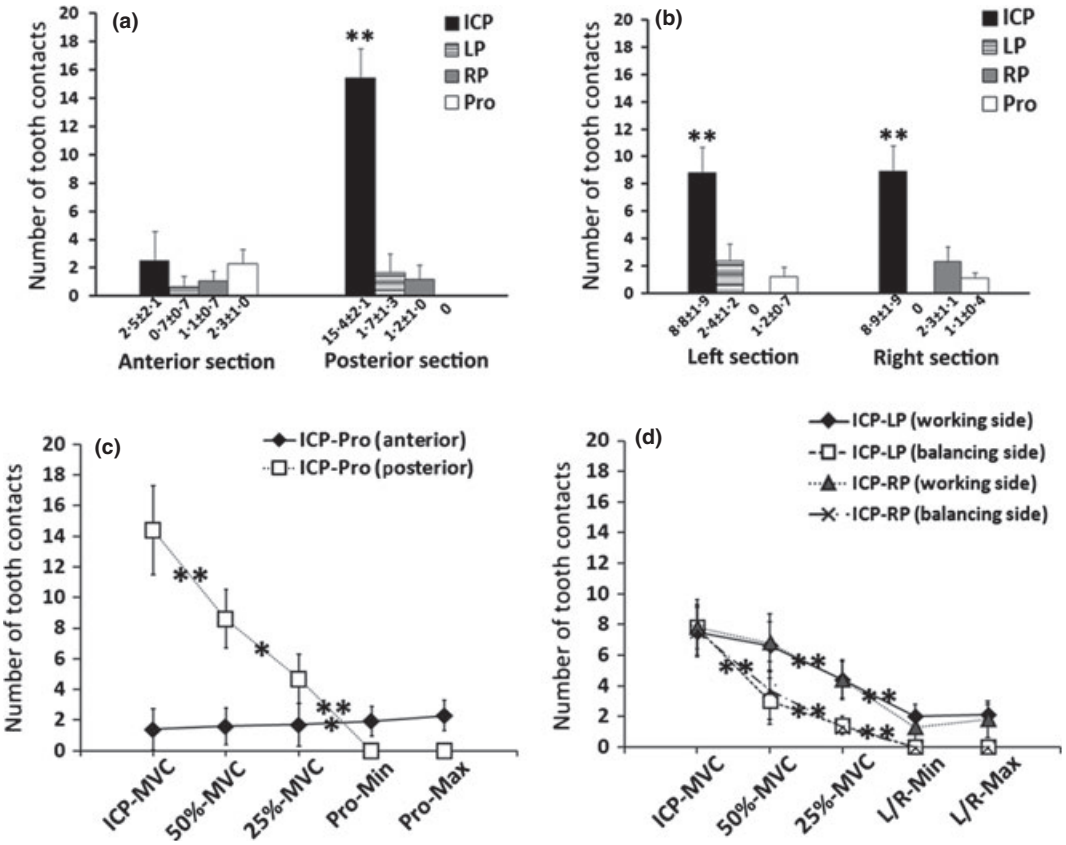

Fig. 4. Mean (± s.d.) of the occlusal contact number in ICP, Pro, LP and RP in anterior and posterior sections (a, c) and in left and right sections (b, d) of the arch during maximal voluntary clenching (a, b) and during protrusive (c) and right or left lateral extensive movements (d) from the maximum intercuspal position. The numbers at the bottom in the x-axis of 'a' and 'b' mean the values of the number of tooth contacts. Other abbreviations as in Fig. 3.

from 25%-MVC to MVC ($P > 0{\cdot}05$) during fast MVC in Pro/LP/RP (Fig. 5).

There were significant decreases in normalised SEMG activity of bilateral TA and MM from ICP-MVC to Pro-Min or L/R-Min, which was the lowest recording value ($P < 0{\cdot}05$). From Pro-Min to Pro-Max, there were increases in bilateral MM ($P < 0{\cdot}05$), but no change in TA ($P > 0{\cdot}05$). From L/R-Min to L/R-Max, there were increases on working-side TA and balancing-side MM ($P < 0{\cdot}05$), but no changes on working-side MM and balancing-side TA ($P > 0{\cdot}05$) (Fig. 6).

Discussion

The present hypothesis was rejected because during fast MVC from rest to ICP, there was a decrease but not an increase in SEMG activity when the bite force increased from 50%-MVC to MVC and the occlusal contact number was increased and more bilaterally distributed. The results also showed that with voluntary clench in ICP from light to heavy, the posterior arch sustained the predominant percentage of bite force with increasing occlusal contacts, resulting in a central posterior location of the centre of bite force (COF). This was accompanied by symmetrical SEMG activity in TA and MM. The discrepancy of the peak values between biting force and SEMG values during the process of fast MVC in ICP should be a unique finding identified in this study using the simultaneous recording system of BioEMG III (Bioresearch Associates, Inc.) and T-Scan – the biosensor was only 0·1 mm thick.

Bite force originates from contraction of the elevator muscles driven by central nervous system during voluntary clench and reflexively adjusted by the peripheral inputs (21, 22) to avoid damage to the structures involved (23–25). Thus, it is generally accepted that muscle activity is affected by the loading stimulus of tooth (26) and dynamic sensitivity of the periodontal receptors to the changes in bite force

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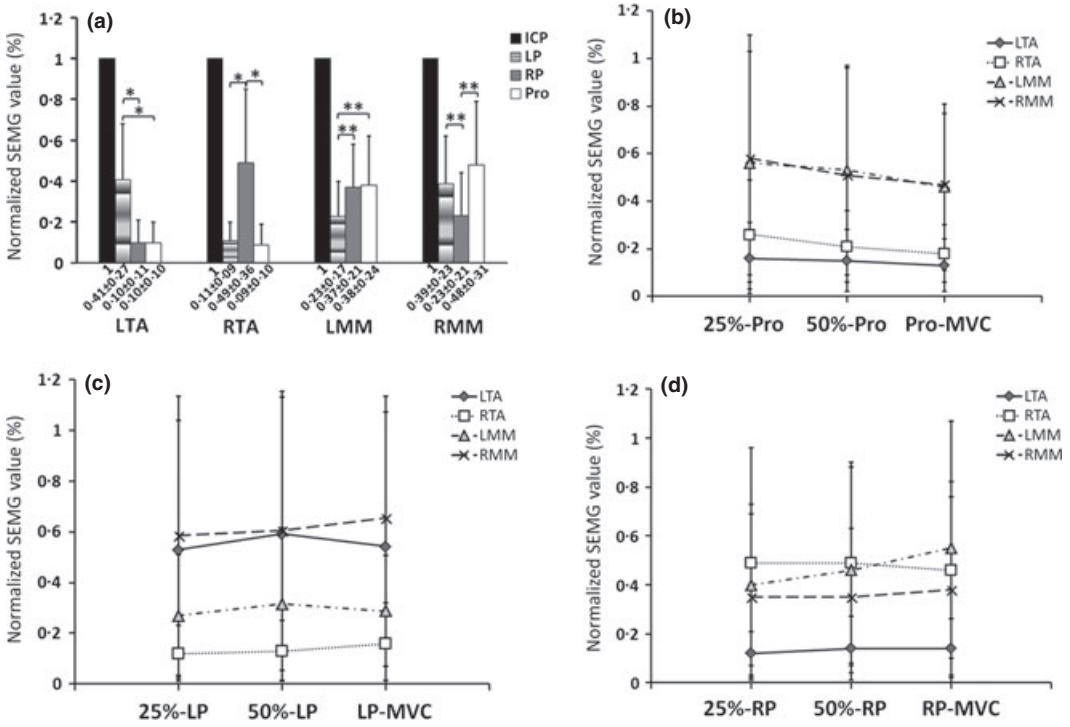

Fig. 5. Mean (± s.d.) of the normalised SEMG activity values of bilateral anterior temporalis (TA) and masseter muscles (MM) during maximal voluntary clenching (MVC) in different mandibular positions. The numbers at the bottom in the x-axis of 'a' mean the normalised SEMG values in different muscles. Other abbreviations as in Figs. 2 and 3.

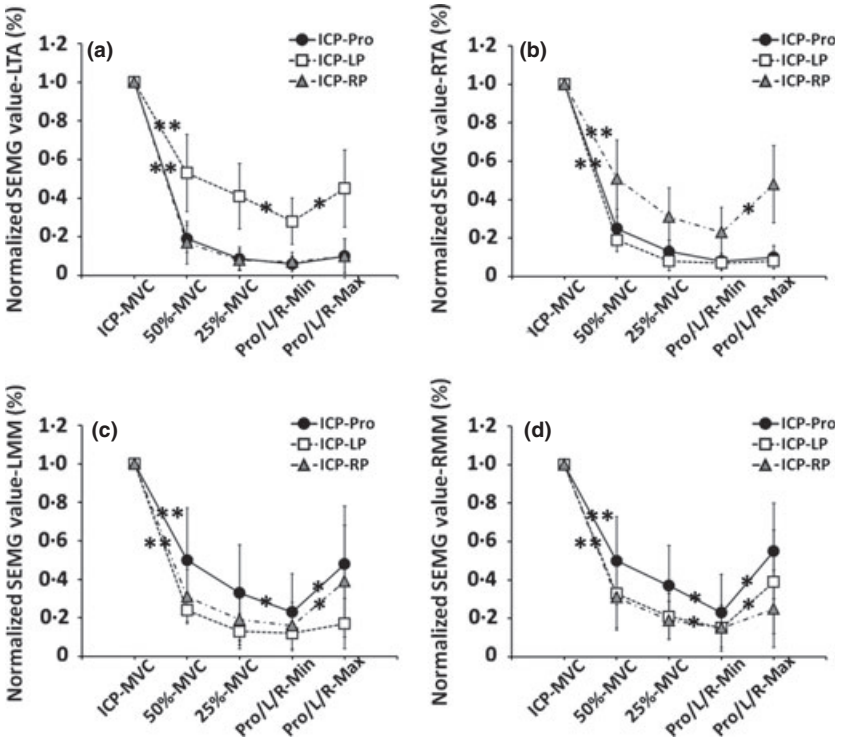

Fig. 6. Changes in mean (± s.d.) of the normalised SEMG activity values of anterior temporalis (a, b) and masseter muscles (c, d) during protrusive and lateral excursive movement from maximum voluntary clenching in ICP. Abbreviations as in Figs. 2 and 3.

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(22). This mechanism is well indicated by that fact that there is an increase in the maximum bite force under periodontal anaesthesia (27). In addition, when a harmful high level of occlusal force is suddenly developed, an unloading jaw reflex will be evoked, through masseteric inhibition, to prevent overload on dental, periodontal tissues or temporomandibular joints (28). This agreement is also achieved by the present data that during movements from ICP to edge-to-edge positions, protrusively or eccentrically, the SEMG values decreased with decreasing bite force and tooth contact numbers.

However, reduction in the SEMG activity is not the all of the elevators' responses to the peripheral inputs in consciousness. Reflex changes in elevator muscle activity occurs with maximum clenching at the edge-to-edge positions and with the SEMG activity of the masseter muscle on the balancing-side being higher than the working-side during MVC with unilateral occlusal support (10, 11). It seems that some subsets of SEMG activity served to stabilise the mandibular bone during clenching. The 'stabilising efforts' of the elevator muscles could not, by the methods now known, be split out from the summary of SEMG activity. However, this 'stabilising' activity of the elevators was identified in this study, as the highest SEMG activity preceding the peak of bite force from light to heavy clenching in ICP. That is, before reaching a stable ICP and generating a high bite force, the elevators adjust their output controlled by the nervous system to direct a precise movement of the mandible. When the number of the occlusal contacts increased to the bilateral maximum of ICP, a balanced jaw position that has a 'bilateral, simultaneous, anterior, and posterior occlusal contact of teeth' (13), a powerful clenching is suitable. Here, the requirement for moving the mandible to a stable position is reduced and then leads to a decreasing of the elevator muscles activity. The supportive results are that when the same voluntary clenching, from light to heavy, were applied in edge-to-edge positions, anteriorly or laterally, where the occlusal contact numbers were far more less than the maximum and the mandible was instable compared to that in ICP, the levels of SEMG activity did not show a significant decrease during MVC. Clearly, unlike the jaw-opening reflex that controls biting force by reducing the jaw-closing muscle activity, a decreased jaw-closing muscle activity was associated with an increased bite force when a stable occlusion was achieved during fast MVC in ICP. This result demonstrates the high efficiency of the elevator muscle contraction, meaning a high level of bite force with a lower level of elevator SEMG activity. Only with a computerised dynamic recording system, could this fast MVC data be identified and analysed within the period of $1{\cdot}55 \pm 0{\cdot}84$ s as found in this study.

The present study has its limitations. First, although the relative force provided by the T-scan III system (Tekscan, Inc.) produces sufficient reliability for bite force comparison within subjects, absolute bite force recording should be made for comparison between subjects. Second, the set-up of simulated physiological occluding function is complex. In the present study, the subject was asked to clench and grind that cannot simulate general function, such as that during chewing. Third, it should be recognised that placement of the sensor between the teeth, although its thickness was only 60–100 μm thick (6, 19), could alter dental proprioception (29) and influence activation of elevator muscles. Finally, on three occasions in one volunteer, the sensor was penetrated during biting tasks, which may be caused by sharp cusps. The tasks have to be repeated with a new sensor re-installed until no further damage to the sensor occurred. Furthermore, a larger number sample of subjects would improve the data. Even more, we cannot explain why during MVC in either ICP or anterior and lateral edge-to-edge positions, the SEMG activity at 25% level was not lower than 50% or maximum (that means MVC) level (Fig. 5). It seems that a high level of the jaw-closing muscle SEMG values during fast MVC is 'setup' at as early as the initial stage of biting. The investigation in this field is attractive.

In summary, the present work, using the simultaneous recording systems for occlusal contacts, bite force and SEMG signals, concluded that the maximum of bite force could be achieved with a submaximal level of jaw-closing muscle SEMG activity in the well-balanced maximum intercuspal relation with the maximum number of occlusal contacts. This means stable tooth contacts in ICP results in utilities of the jaw elevators during maximal voluntary clenching. According to this principle, a high value of the ratio of SEMG to biting force during fast MVC from rest to ICP might be an indication of lower efficiency of the elevators muscles, which may require extra muscle activity to maintain a particular bite force level.

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